



A Commonality Measure for Overlapping Clusters Detections

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What a cluster is?

- There is no universally accepted definition for cluster in graphs.
- Graph partitioning problem. Dozens of algorithms and techniques.
- Sets of nodes that the algorithm finds are then called “clusters,” “communities,” “groups,” “classes,” or “modules”.
- This is Ok! The applications are context-dependent.
- Do we need one more approach? If it has a use case!

Community vs. Cluster

- Two main approaches for network partitioning: by whole network analysis or by local data.
- **Term cluster** is suitable for global approach, when clusters have been recognized by comparison properties of different parts of network.
 - Modularity
 - Betweenness centrality
- **Term community** is suitable for local approach, when a community have been recognized without full network analysis.
 - Clique percolation
 - Label propagation

Natural Community!!

Some intuitive understanding of **overlapping communities** can be derived from social networks...



- Each node “knows” his community’s members: local property.
- Each node may belong to more than one community
- **Topologically:**
two nodes “**surely**” belong to the same community if they have a significant number of common neighbors.

Nodes Commonality

- $N(i)$ is the neighborhood of a node i
- $commonality(N(i), N(j))$ is a function of two nodes to quantify the status of their common neighbors.
- Exists a threshold c_0 , if $commonality(i,j) > c_0$ nodes i,j “for suer” belongs to the same community.
- For different type of networks, $commonality$ may be different functions
- Commonality may be calculated by a pretrained neural network as a probability to be members of the same community.

Commonality - Jaccard coefficient

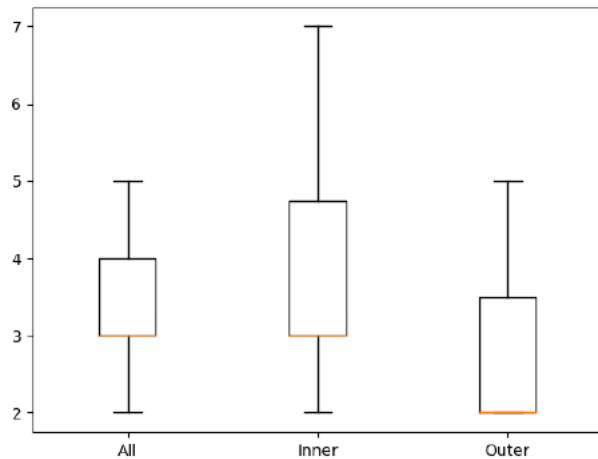
$$c(i, j) = \frac{|N(i) \cap N(j)|}{|N(i) \cup N(j)|}$$



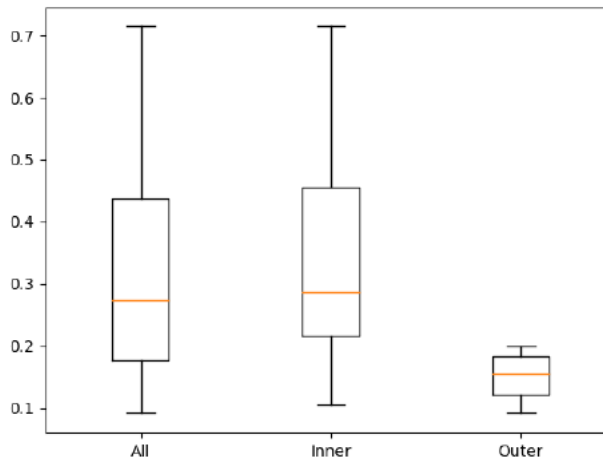
- *commonality* $c(i, j)$ of two nodes i and j is a fraction of common neighbors
- The simplest, but may be not the best
- $c(\text{pink}, \text{green}) = 2/5$, $c(\text{pink}, \text{blue}) = 3/4$

commonality concept test

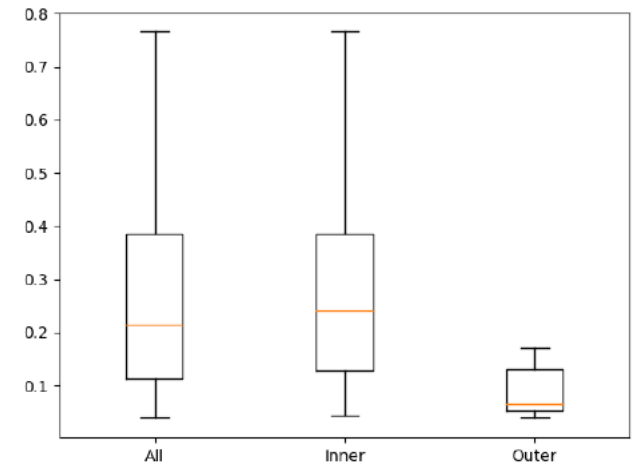
- calculate the commonality for different real-world networks with “ground truth”
- $f1 = |N(i) \cap N(j)|$, $f2 = \frac{|N(i) \cap N(j)|}{|N(i) \cup N(j)|}$, $f3 = \frac{|N(i) \cap N(j)|^2}{|N(i) \cup N(j)|}$
- For Zachary’s Karate Club commonality distribution:



f1



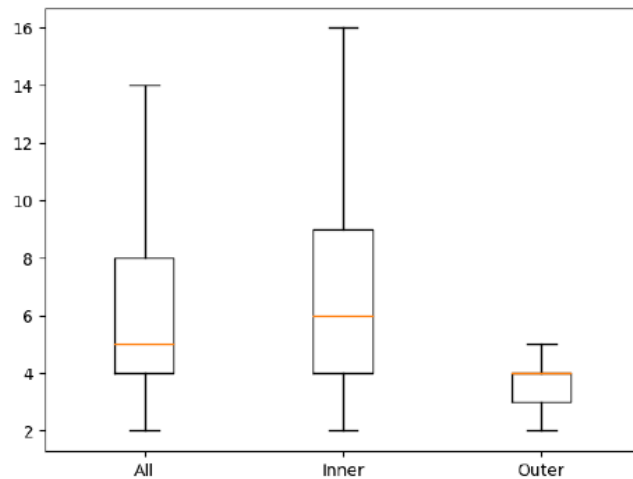
f2



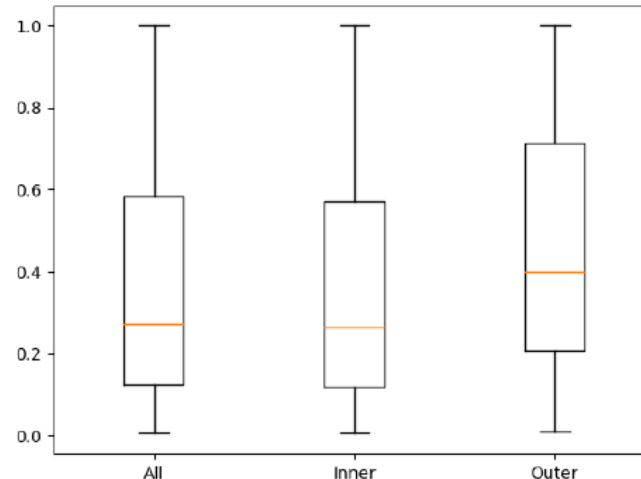
f3

commonality distribution

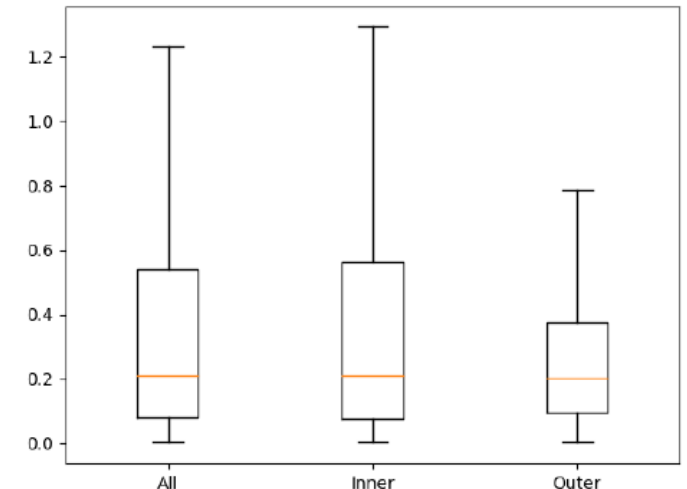
- For DBLP a co-authorship network Jaewon Yang and Jure Leskovec. “Defining and evaluating network communities based on ground-truth”. In: Proceedings of the ACM SIGKDD Workshop on Mining Data Semantics. 2012, pp. 1–8.
- $f1 = |N(i) \cap N(j)|$, $f2 = \frac{|N(i) \cap N(j)|}{|N(i) \cup N(j)|}$, $f3 = \frac{|N(i) \cap N(j)|^2}{|N(i) \cup N(j)|}$



f1

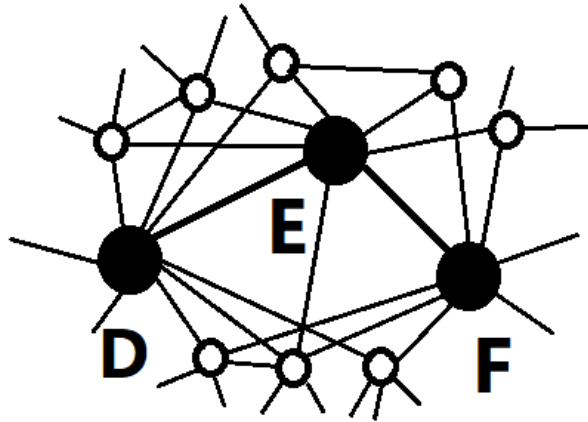
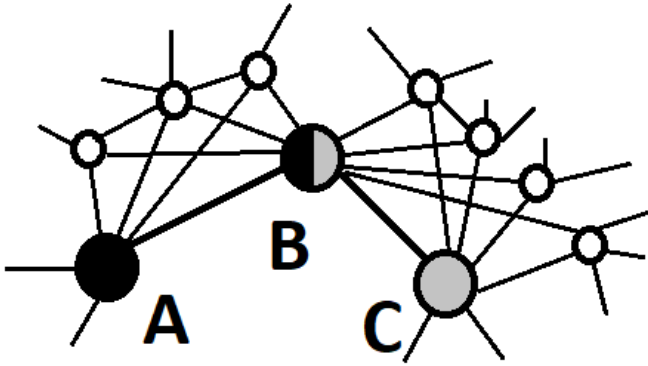


f2



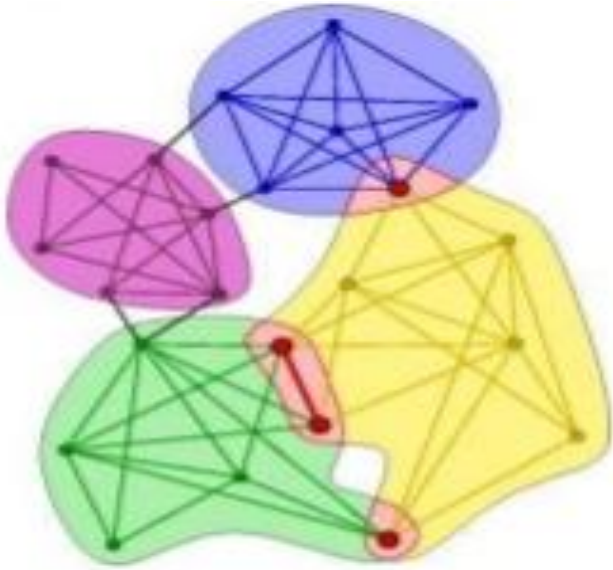
f3

Hypergraph constructing



- A link (i,j) is inside a community “for sure” (inner link) if $c(i,j) > c_0$
- (A,B) , (B,C) , (D,E) , (E,F) are "for sure" inner links. $c(A,B)=3/7$, $c(B,C)=4/10$, $c(A,C)=1/13$, $c(D,E)=4/11$, $c(E,F)=3/11$, $c(D,F)=4/13$.
- Three nodes having $c(i,j) > c_0$ are a hypernode.
- (D,E,F) is a hypernode. (A,B,C) is not.
- Two hypernodes having two nodes in common are connected by hyperlink.

Natural Community - definition



- Natural community is a set of nodes belonging to a connected component of the hypergraph.
- Natural communities are overlapping.

Algorithm for a natural community detection

Input: network $G=(V,E)$, threshold value c_0 ;

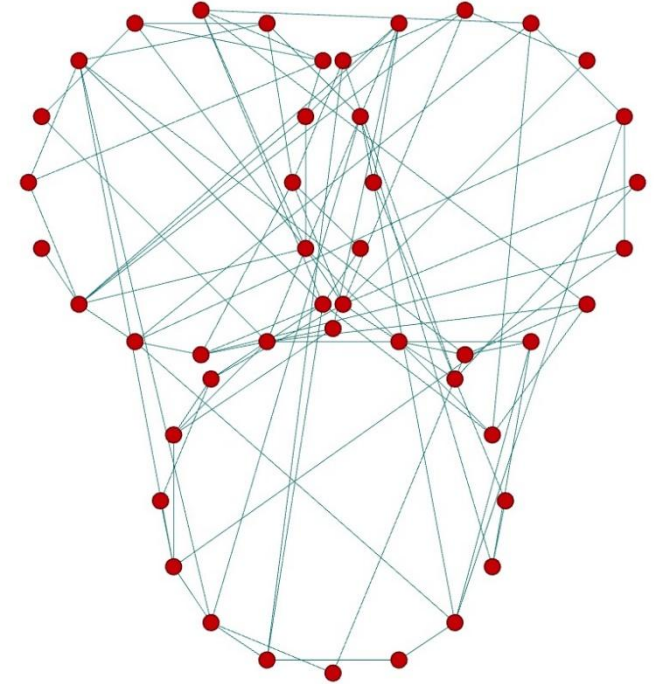
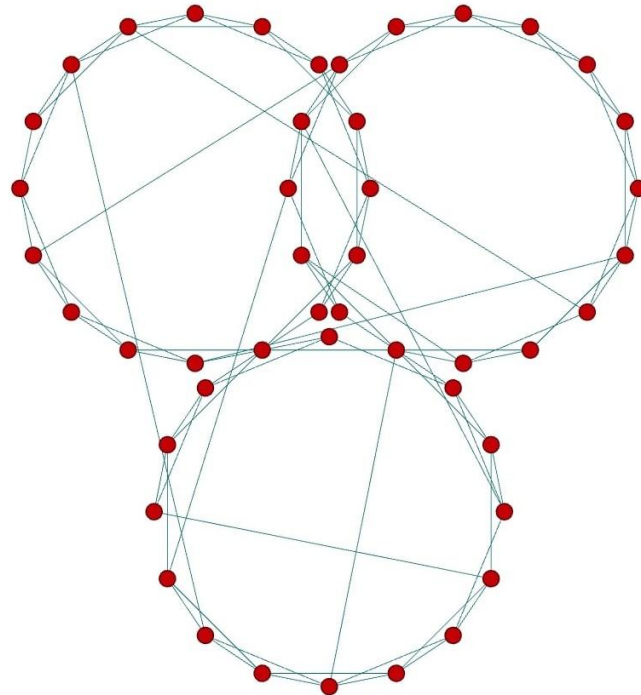
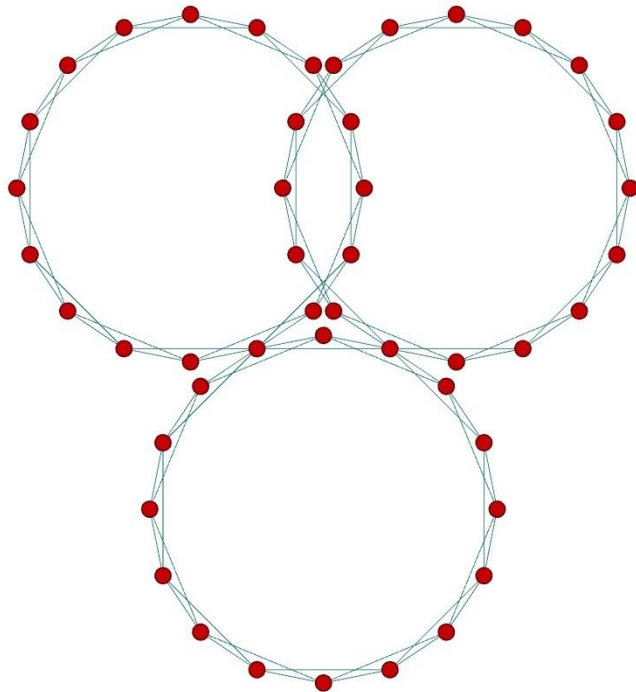
Output: A community $C \subset G$.

1. Start with arbitrary node v , $C=\{v\}$;
2. Loop for each $w \in N(v)$
 if $c(v,w) > c_0$ put (v,w) into queue Q ; break;
3. loop while Q is not empty
 - a. pop (v,w) from Q ;
 - b. loop for each $u \in N(w)$
 if $(c(w,u) > c_0$ and $c(w,v) > c_0$ and $u \notin C)$
 put (w, u) into Q ; add w and u to C ;

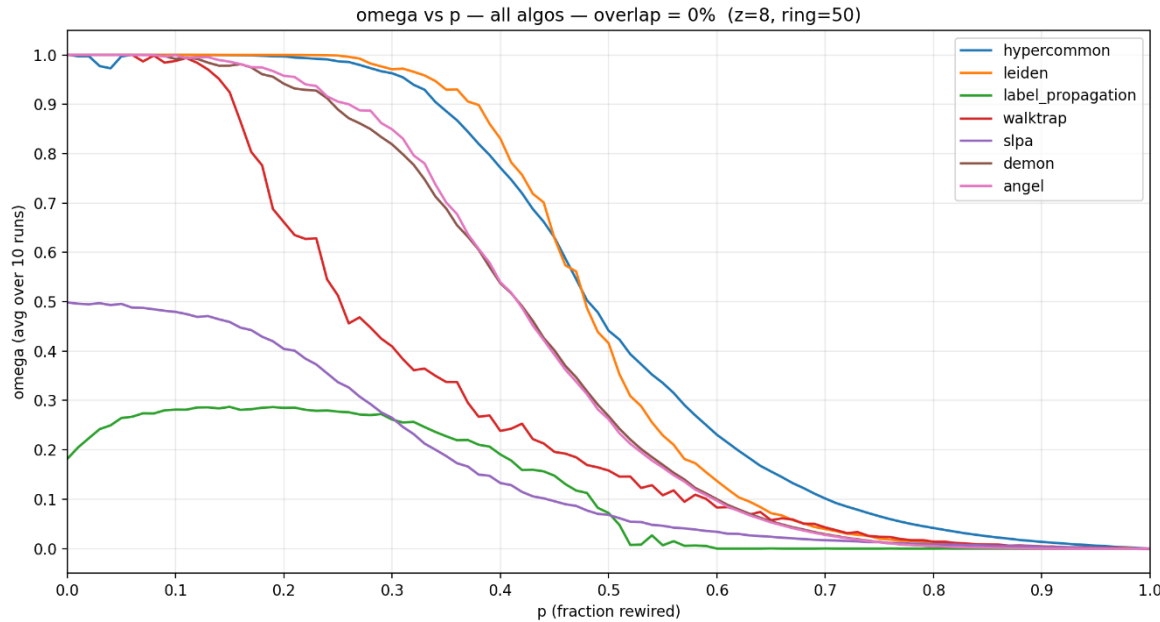
$N(v)$ is set of the
node v neighbors

Synthetic data Test Case: Small World Graph Extension

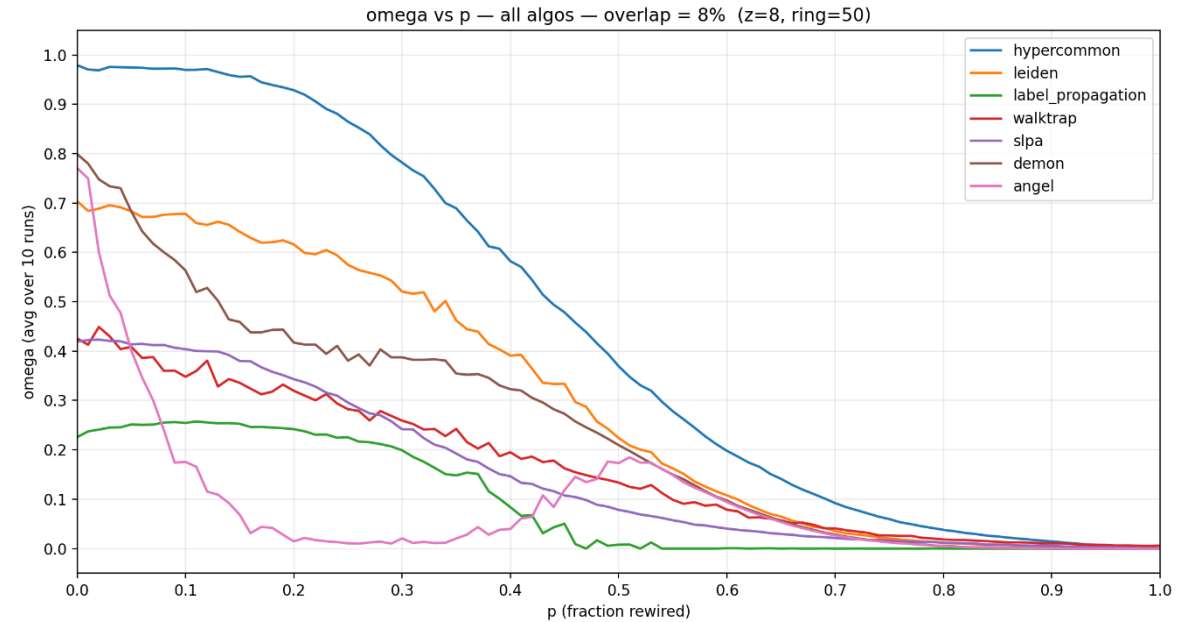
- Collection of ring lattices with randomly reconnected P_{in} links inside the ring lattice and P_{out} reconnected links between the rings
- For overlapping case P_{over} randomly chosen are common for rings nodes.



Simulation for the Test Case



No overlapping



8% overlapping

Fraction of correct recognized communities vs. level of randomization.
Different popular algorithms. **Blue line** – by hypergraph constructing.

Conclusion

- **Commonality** quantifies the potential of two nodes to belong to the same community, based on their shared neighbors.
- A **hypernode** is defined as a set of three nodes having high mutual commonality.
- A **hyperlink** exists between two hypernodes having two nodes in common.
- A **Natural Community** is a connected component of the hypergraph.
- **The algorithm** developed from these definitions is straightforward, utilized local data, efficient and effective. It also exhibits stability in the face of random link perturbations.

Thank you.

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